

BELEM, Brazil

WMO: 821930

Lat: 1.379S

Lon: 48.476W

Elev: 17

StdP: 101.13

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.8	22.9	19.9	14.6	28.4	20.5	15.2	27.6	9.8	27.5	8.4	27.9	2.3	100	0.352

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	7.9	33.2	25.8	33.0	25.8	32.2	25.7	28.1	30.4	27.7	30.2	27.4	30.1	3.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.1	29.5	27.1	22.8	29.4	26.8	22.5	29.1	90.1	30.8	88.7	30.3	87.0	30.4	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.0	6.1	21.8	34.5	0.6	0.9	21.3	35.1	21.0	35.6	20.6	36.1	20.2	36.7		
(5)			20.7	28.4	1.1	1.0	19.9	29.2	19.3	29.8	18.7	30.4	17.9	31.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	27.2	27.0	27.0	27.3	27.7	27.8	27.9	27.9	27.8	28.0	27.8	(6)	
(7)		DBStd	0.94	1.08	1.17	1.04	0.99	0.92	0.66	0.72	0.75	0.69	0.66	0.74	0.86	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6431	534	475	528	521	549	536	553	554	536	553	541	551	(10)
(11)		CDD18.3	3389	276	242	269	271	291	286	295	296	286	295	291	292	(11)
(12)		CDH23.3	30822	2294	1863	2107	2231	2567	2673	2807	2843	2761	2951	2997	2728	(12)
(13)		CDH26.7	10367	653	477	516	594	817	991	1085	1084	1031	1094	1116	911	(13)
(14)	Wind	WSAvg	2.9	2.6	2.6	2.5	2.4	2.5	2.8	3.0	3.2	3.4	3.3	2.9	(14)	
(15)	Precipitation	PrecAvg	3034	373	384	466	408	310	194	176	157	119	112	110	225	(15)
(16)		PrecMax	3666	515	564	629	547	463	294	312	233	194	206	276	440	(16)
(17)		PrecMin	2551	237	256	253	289	180	60	64	76	38	14	21	68	(17)
(18)		PrecStd	324	74	76	82	67	69	52	54	42	39	54	58	95	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	33.1	32.7	33.0	33.2	33.2	33.2	34.0	33.8	33.6	33.9	33.2	(19)
(20)			MCWB	26.4	26.6	26.5	26.8	26.6	25.7	25.3	25.5	26.0	25.7	26.0	25.8	(20)
(21)		2%	DB	31.9	31.7	31.2	31.8	32.2	32.4	32.8	32.9	32.8	32.6	32.8	32.2	(21)
(22)			MCWB	26.2	26.1	26.3	26.7	26.2	25.5	25.2	25.6	25.7	25.7	25.7	25.8	(22)
(23)		5%	DB	30.9	30.6	30.2	30.9	31.8	32.0	32.1	32.0	31.9	31.9	31.9	31.2	(23)
(24)			MCWB	26.2	26.1	26.2	26.5	26.2	25.4	25.2	25.6	25.7	25.7	25.8	26.0	(24)
(25)		10%	DB	30.0	29.6	29.2	29.9	30.8	31.2	31.2	31.2	31.0	31.0	31.0	30.8	(25)
(26)			MCWB	26.0	25.9	25.8	26.2	26.2	25.6	25.4	25.8	25.9	26.0	26.0	26.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.0	28.0	27.9	28.4	28.2	27.7	27.6	28.0	28.0	28.1	28.1	28.2	(27)
(28)			MCD B	30.1	30.4	29.9	30.6	30.7	30.4	30.4	30.5	30.4	30.3	30.4	30.3	(28)
(29)		2%	WB	27.4	27.2	27.3	27.6	27.5	27.1	27.1	27.3	27.3	27.4	27.5	27.5	(29)
(30)			MCD B	29.8	29.7	29.4	29.9	30.3	30.4	30.4	30.5	30.4	30.3	30.3	30.2	(30)
(31)		5%	WB	26.9	26.6	26.7	27.1	27.0	26.6	26.6	26.7	26.8	27.0	27.0	26.9	(31)
(32)			MCD B	29.5	29.0	28.9	29.3	30.0	30.0	30.1	30.3	30.3	30.3	30.3	30.0	(32)
(33)		10%	WB	26.3	26.1	26.2	26.5	26.5	26.1	26.1	26.3	26.3	26.4	26.5	26.4	(33)
(34)			MCD B	28.7	28.3	28.2	28.7	29.3	29.6	29.8	29.9	29.8	29.8	29.9	29.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.8	6.5	6.3	6.5	7.1	7.9	8.2	8.2	8.1	7.9	7.5	(35)	
(36)			MCDBR	7.6	7.6	7.1	7.4	8.0	8.4	8.6	8.7	8.7	8.6	8.4	8.0	(36)
(37)			MCWBR	3.3	3.2	3.2	3.2	3.2	3.1	3.3	3.6	3.5	3.6	3.4	3.4	(37)
(38)		5% WB	MCDBR	7.1	6.9	6.6	6.9	7.4	8.0	8.2	8.3	8.4	8.2	8.0	7.8	(38)
(39)			MCWBR	3.5	3.4	3.3	3.4	3.3	3.4	3.5	3.8	3.8	3.8	3.5	3.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.416	0.422	0.425	0.403	0.410	0.405	0.416	0.435	0.494	0.489	0.490	0.467	(40)	
(41)		taud	2.486	2.470	2.467	2.551	2.508	2.518	2.473	2.395	2.196	2.218	2.218	2.310	(41)	
(42)		Ebn at Noon	917	912	901	899	869	860	856	859	828	842	842	864	(42)	
(43)		Edh at Noon	114	117	117	104	105	102	108	120	151	149	148	135	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.65	4.35	4.49	4.70	4.97	5.36	5.62	5.93	5.97	5.77	5.56	4.99	(44)	
(45)		RadStd	0.29	0.30	0.31	0.25	0.34	0.20	0.16	0.14	0.17	0.24	0.21	0.30	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page